

Workshop on Human Motion Challenges in Real-World and Clinical Settings

(MoCha) @ ECCV 2026

<https://mocha.care-pd.ca/>

👉 Each team must submit the [team registration form](#). If your team needs to update any information after submission, please contact us by email (mocha.eccv@gmail.com). Please note that team registration information is **required** for award eligibility and leaderboard verification.

Benchmark and Challenge on Parkinsonian Gait

MoCha2026 evaluates methods for predicting MDS-UPDRS gait severity classes from SMPL motion sequences.

Participants submit code and model weights; CodaBench runs the submitted code on hidden motion files and updates the leaderboard.

Task

For each motion sequence, predict the gait severity class:

0, 1, 2, or 3

Metric

Submissions are ranked by overall Macro F1. Macro precision, macro recall, accuracy, and Quadratic Weighted Kappa are also reported. Quadratic Weighted Kappa is included because UPDRS classes are ordinal, so farther mistakes are penalized more strongly.

Awards

- **First Place Award (€1,000)**
- **Second Place Award (€500)**

Sponsored by [Machine Medicine Technologies](#)

Submission Processing

Submitted methods are processed in batches approximately every 6 hours, after which the leaderboard will be updated. Submissions are capped at a maximum of 3 attempts per participant per day.

Provided Resources

Please visit [MoCha_baseline_bundle GitHub](#) for an example submission script.

Contact: Please use the forum on the challenge page for technical discussions, or contact the MoCha organizing committee via the official website: <https://mocha.care-pd.ca/>, or directly contact the challenge organizers by email (mocha.eccv@gmail.com) if you have doubts or any question.

Final Submission and Winner Selection

A panel of experts will review the entries. Rankings are primarily based on the quantitative F1 metric defined in the leaderboard. The judges' decisions are final. In the event of a tie, the submission with the earlier timestamp on the server will prevail.

The final submission should be emailed to the organizers (mocha.eccv@gmail.com) **before the challenge phase ends**.

The email title should be:

[MoCha2026] - [TEAM_NAME]

The body of the email should include the following:

- Challenge full name: MoCha 2026 - Challenge Final Submission
- Team name
- Team leader's name and email address
- Rest of the team members (if applicable)
- Affiliations of the team member(s)
- Team name and username(s) on the MoCha 2026 CodaBench server
- Executable/source code attached or download links

The executable/source code should include trained models or necessary parameters to ensure reproducibility.

Note: Submitted codes will not be saved on the evaluation servers. Final submission codebases provided to the organizers will be used only for reproducibility checks and winner verification, and will not be shared publicly or with other participants at any stage.

Data & Evaluation

Training Data

Participants should use the CARE-PD dataset for training.

Resources

Resource	Link
 CARE-PD Website:	https://neurips2025.care-pd.ca/
 Hugging Face Dataset:	https://huggingface.co/datasets/vida-adl/CARE-PD
 Canonicalized SMPL Files:	Canonicalized SMPL pickles
 Submission Format Example:	MoCha Baseline Bundle

CARE-PD is a multi-site, anonymized clinical dataset for Parkinson's disease gait assessment. Participants may use data from any available training sites to develop their models. Submitted models will be evaluated on held-out data from different clinical sites to assess their ability to generalize across real-world site variations.

The hidden test benchmark follows the same data format as the released [canonicalized SMPL pickles files](#).

Input Format

The hidden evaluation set contains canonicalized SMPL walking sequences from Parkinsonian gait datasets. Labels are kept private on our server.

Each input file, in both the training and hidden test sets, is a Python pickle containing a nested dictionary:

```
data[subject_id][walk_id] = {  
    "pose": np.ndarray, # shape (T, 72), SMPL axis-angle pose  
    "trans": np.ndarray, # shape (T, 3), global translation  
    "beta": np.ndarray, # shape (1, 10), all zeros for privacy  
    "fps": int,  
}
```

The canonicalized SMPL coordinate convention is:

x = lateral

y = up

z = forward

The first frame starts near the ground-plane origin in x and z, and the body is placed on the floor plane. Participants should still use the provided fps field if their method is frame-rate sensitive.

Note:

- For privacy, all SMPL shape coefficients (beta) are set to zero in the canonicalized files.
- Preprocessing should also handle FPS conversion to the rate expected by each model. The original data collection frame rate is stored inside each pickle entry under the fps key.

Labels

The target label is the MDS-UPDRS gait severity class:

0, 1, 2, 3

Benchmark test labels are hidden from participants during evaluation.

Metrics

The primary leaderboard metric is:

Macro F1

The leaderboard also reports:

Macro precision

Macro recall

Accuracy

Quadratic Weighted Kappa (QWK)

QWK is included as an ordinal-aware metric: predicting a far-away class, such as 0 instead of 3, is penalized more strongly than predicting a neighboring class.

Missing predictions and invalid predictions are treated as incorrect.

Submission

Submit a zip file with a run.py file at the root of the zip.

Your run.py must define:

```
def predict(data: dict) -> dict:
```

```
...
```

CodaBench will call your function with the unseen data:

```
data[subject_id][walk_id] = {  
    "pose": np.ndarray, # shape (T, 72)  
    "trans": np.ndarray, # shape (T, 3)  
    "beta": np.ndarray, # shape (1, 10)  
    "fps": int,  
}
```

Your function must return:

```
predictions[subject_id][walk_id] = predicted_class
```

where predicted_class is an integer in {0, 1, 2, 3}.

Additional folders may be included in your zip:

- Model weights
- Helper modules imported by run.py
- Any preprocessing files needed by your model

Participants do not need to write prediction files manually. The ingestion program serializes the returned prediction dictionary for the scorer.

Submission Processing

Submitted methods are processed in batches approximately every 6 hours, after which the leaderboard will be updated.

Submissions are capped at a maximum of 3 attempts per participant per day.

To submit your code go to **My Submissions** and upload your zip file. After a few hours you will be able to see your final score and you can add your score to the leaderboard by clicking on the green window icon in the **Actions** field name as **Add to Leaderboard**. Checkout the Leaderboard on the **My results** tab.

Docker Environment

This challenge uses a Docker-based evaluation environment. Do not rely on installing packages at submission time. Your submission should run with the packages available in the official Docker image, and all required model weights or pure-Python helper code should be included in your submission zip.

The zip root must contain run.py. Other files can be organized in folders.

For a complete example, see:

https://github.com/TaatiTeam/MoCha_baseline_bundle

Timeline

All deadlines are end-of-day Anywhere on Earth (AOE).

Challenge Timeline

- CARE-PD data is already released for model development and training.
- Challenge leaderboard submission opens: **May 31, 2026**.
- Challenge leaderboard submission closes: **August 15, 2026**.
- Winner announcement on the workshop website: **August 18, 2026**.
- MoCha @ ECCV 2026 workshop: **Sep 9**, Malmo, Sweden.

Challenge submissions are evaluated automatically by CodaBench on the hidden test benchmark. The top-performing teams will be selected according to the official leaderboard rankings.

Paper Submission

Challenge participants may also submit a corresponding paper. Paper submissions are handled separately from the CodaBench leaderboard and must follow the official MoCha workshop paper deadlines.

MoCha accepts papers in two tracks:

Track 1: Proceedings Track

This track is for novel full papers that will appear in the ECCV workshop proceedings if accepted.

- Submission deadline: **July 8, 2026 (23:59 AOE)**.
- Notification date: **July 25, 2026**.
- Camera-ready deadline: **August 3, 2026 (23:59 AOE)**.
- Format: full papers only, up to 14 pages in ECCV style.

Track 2: Non-Proceedings Track

This track is non-archival and is intended for a broader range of work, including novel or previously published work.

- Submission deadline: **July 23, 2026 (23:59 AOE)**.
- Notification date: **August 15, 2026**.
- Format: extended abstracts, short papers, or full papers in ECCV style.

Accepted papers from both tracks may be invited for poster, spotlight, or oral presentation at the workshop. The challenge leaderboard remains open until the challenge submission deadline, even if a team also submits a paper.

For the latest workshop information, please see:

<https://mocha.care-pd.ca/>

Terms and Conditions

By registering for the challenge, you agree to the following terms.

Data Privacy

- Do not attempt to re-identify any subjects from the motion data.
- Do not download, scrape, or export hidden test data from CodaBench.
- Use the data only for this scientific challenge.

Submission Rules

- Submit code and optional model weights in a zip file.
- The zip root must contain run.py.
- run.py must define predict(data) -> dict.
- Submissions must run inside the official Docker-based evaluation environment.
- Participants may not use multiple accounts to evade submission limits.

Liability

The organizers provide the data and evaluation server as-is and make no warranties regarding completeness or availability.

Disqualification

Participants who violate data privacy, submission rules, or CodaBench platform rules may be disqualified.

Chasuite

[Competitions v1.6ChahubChagrade](#)

About

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CodaBench

Download	Phase	Task	Type
	Evaluation Phase	Evaluation Task	Solution